1. **Course Plan**

2. **AWS SageMaker**
   - Introduction

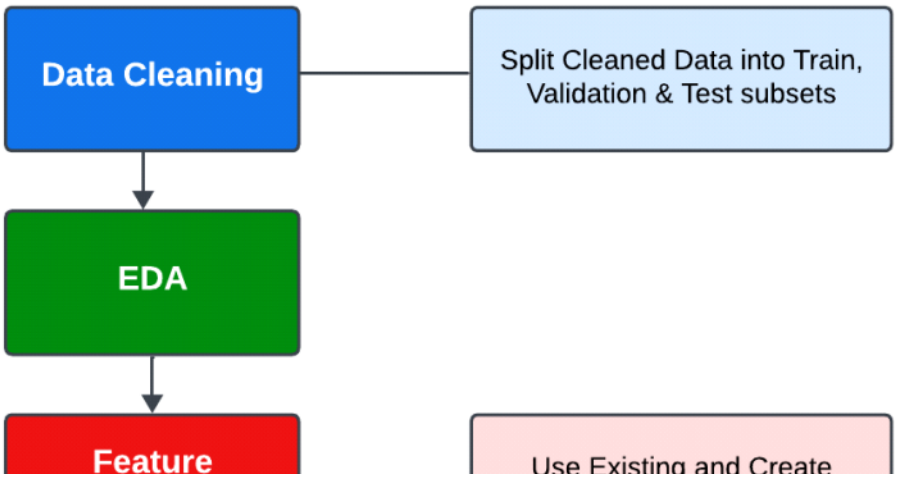
3. **AWS services:**
   - IAM
   - EC2
   - S3

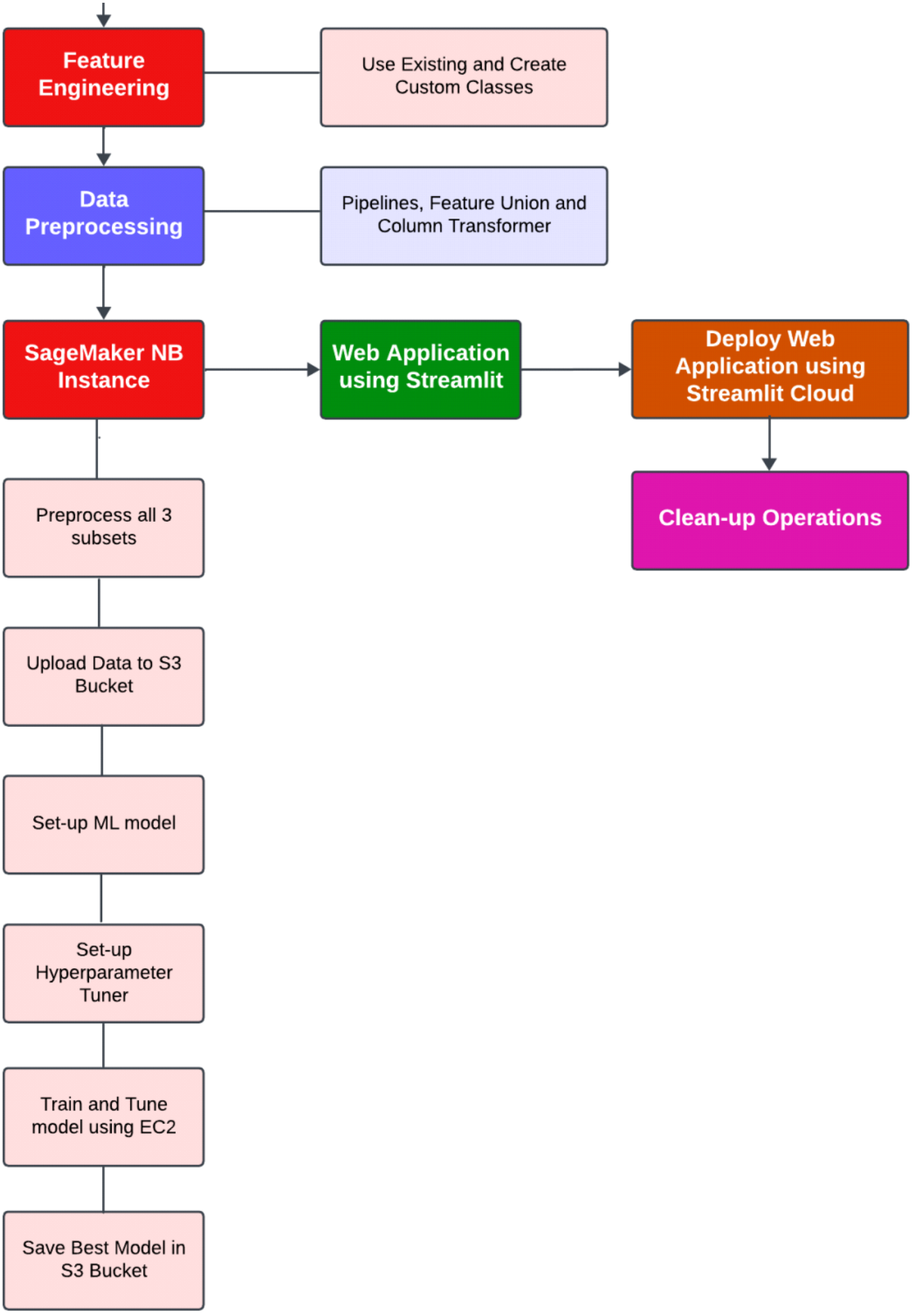
4. **SDK (Software Development Kit)**
   - Definition
   - Advantages
   - Boto3

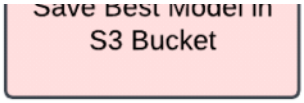
5. **AWS Free Tier Plans**

6. **Create AWS account**

7. **Project Workflow**



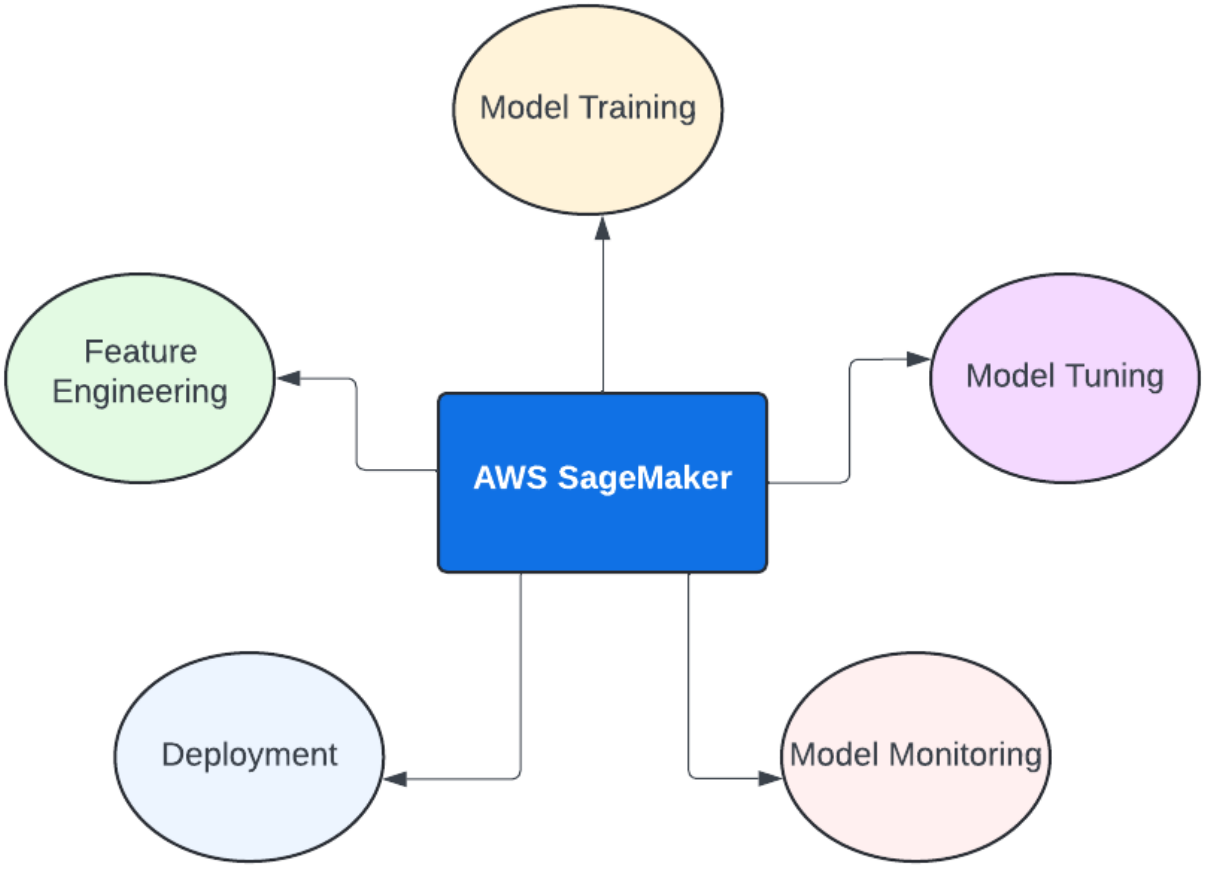

Feature Engineering
Use Existing and Create Custom Classes
Data Preprocessing
Pipelines, Feature Union and Column Transformer
SageMaker NB Instance
Web Application using Streamlit
Deploy Web Application using Streamlit Cloud
Clean-up Operations
Preprocess all 3 subsets
Upload Data to S3 Bucket
Set-up ML model
Set-up Hyperparameter Tuner
Train and Tune model using EC2
Save Best Model in S3 Bucket



8. Create SageMaker Notebook Instance

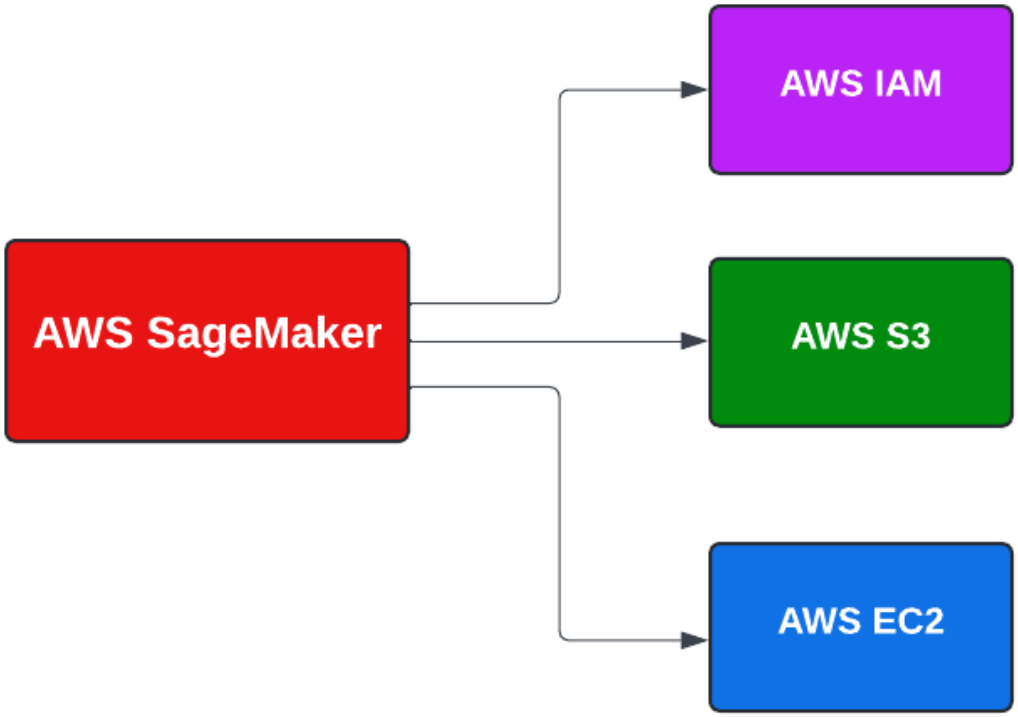
1. Definition

Amazon SageMaker is a fully managed machine learning (ML) service. With SageMaker, data scientists and developers can quickly and confidently build, train, and deploy ML models into a production-ready hosted environment. It provides a UI experience for running ML workflows that makes SageMaker ML tools available across multiple integrated development environments (IDEs).



2. **Advantages**
    - It's a fully-managed service
    - Provides wide range of libraries for ML workflows
    - Provides wide range of algorithms for various problems
        - *Regression*
        - *Classification*
        - *Object Detection*
        - *Forecasting*
        - *Clustering*
        - *Dimensionality Reduction*
        - *Recommendation Systems*
    - One-click deployment
    - Integration with other AWS services
    - Provides notebooks and IDE for development
        - *Classic Jupyter*
        - *Jupyter Lab*
        - *SageMaker Studio (IDE)*
    - Detailed guides and tutorials
    - Huge community and online support

Apart from SageMaker, we'll be using the following services from AWS to implement our End-to-End ML Project:



1. **IAM**
2. **EC2**
3. **S3**

1. Definition

What is an SDK?

A software development kit (SDK) is a set of platform-specific building tools for developers. You require components like debuggers, compilers, and libraries to create code that runs on a specific platform, operating system, or programming language. SDKs put everything you need to develop and run software in one place. Additionally, they contain resources like documentation, tutorials, and guides as well as APIs and frameworks for faster application development.



2. Advantages

- Efficient Development
- Reduced time complexity
- Seamless Integration

- Reduced Costs

3. **Why did we study about SDK?**
   - We'll be using Python to train models
   - Need a way for Python to interact with AWS services

What SDKs does AWS provide?

AWS provides SDKs for many popular technologies and programming languages. They make it easier for you to call AWS services from within your applications in that language or technology. Additionally, AWS also offers SDKs for AWS SaaS offerings so you can use them more efficiently in your code. Some examples include:

- AWS SDK for .NET provides simplified AWS Services through a set of libraries that are consistent and familiar to .NET developers.
- AWS SDK for Python integrates Python applications, libraries, or scripts with AWS services.
- AWS SDK for Ruby eliminates programming intricacies by providing Ruby classes for many AWS services.
- AWS SDK for Rust simplifies the use of AWS services by providing a set of consistent and familiar libraries to Rust developers.
- AWS WorkDocs SDK removes the complexity of building file collaboration and management capabilities into your solutions and applications, providing full administrator and user-level access to Amazon WorkDocs site resources.
- Amazon Chime SDK allows developers to add real-time voice, video, and messaging powered by machine learning into their applications.